

Samiur Rahman

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EDUCATION

Georgia Institute of Technology

Atlanta, Georgia

B.S. in Computer Engineering

May 2026

- **Concentrations:** Distributed System & Software Design and Computing Hardware & Emerging Architecture
- **GPA:** 3.82/4.00, *Dean's List*
- **Related Coursework:** Data Structures & Algorithm; Object-Oriented Programming; GPU Programming, Digital Design; Embedded System Design & Programming; Computer Architecture, Concurrency & Energy in Computation

EXPERIENCE

Batch-88 Foundation

Atlanta, Georgia

Full-Stack Web Developer

July 2024 – Current

- Developing web platform for 5,000+ users; load times optimized by 60% with React.js, Next.js, and Tailwind CSS
- 100% authentication accuracy for 800+ members/admins via Firebase Authentication, OAuth, and RBAC
- Achieved 45% query latency reduction and 70% reduction of manual admin work through Firestore optimization
- Achieved 99.95% uptime and 50% improved media management with Firebase Hosting and Cloudinary integration

Citadel Technologies

Atlanta, Georgia

Software Developer Intern

May 2024 – Aug 2024

- Developed a Flutter mobile app for remote garage control using WebSocket APIs, achieving 95% user satisfaction rate
- Implemented multi-garage and home view support using Provider and RESTful APIs, reducing user setup time by 40%
- Integrated IoT sensors and Firebase Cloud Messaging for real-time alerts, increasing response times by 30%

PROJECTS

CUDA Image Blurring Algorithm for High-Performance Computing

Atlanta, Georgia

GPU Programming Project

Aug 2024 – Sep 2024

- Developed a CUDA-based image blurring algorithm optimized for 4096x4096 images using shared memory, tiling, and texture memory techniques on an NVIDIA H100 GPU with interactive UI, built with SFML library to show performance
- Achieved 100x speedup over CPU, reducing time from 5s to 0.5s, optimizing memory access and thread management
- Minimized memory bandwidth by 60%, with 95% GPU occupancy, enabling efficient pixel-averaging convolution

Top of the Pile

Atlanta, Georgia

ECE Robotics

Jan 2024 – May 2024

- Built an algorithm for a robotic system to identify the topmost part in a pile using a 64x64 image array using C
- Improvised solution using MIPS assembly that analyzes the color composition of parts to determine the sequence in real-time, ensuring top 1% computational and storage efficiency, preventing malfunctioning of robotic assembly
- Increased algorithm accuracy by 30%, reducing assembly errors with precise color-based identification

Galaxy Wars Shooting Game

Atlanta, Georgia

Embedded System Project

Jan 2024 – May 2024

- Programmed a handheld shooter game for the Mbed platform utilizing C++, enhancing game performance by 30%
- Implemented a comprehensive suite of player interactions, including sprite-based graphics, enemy engagements, multiple difficulty levels and a scoring system with status updates, resulting in a 40% increase in player engagement
- Created advanced features such as multiple user access up to 5 different users with alongside password protection

Motor Control Peripheral

Atlanta, Georgia

Digital Design Lab Project

Jan 2024 – Apr 2024

- Led the design of a servo motor control peripheral for SCOMP using VHDL, enabling precise control of 4 motors
- Performed timing analysis for 0.001-degree precision configurable pulse width range and reduced response latency
- Applied configurable pulse width ranges and velocity control, ensuring up to 0.001 degrees precise motor movement
- Simulated and verified designs using ModelSim, achieving 95% accuracy in timing and functionality before testing

SKILLS

Languages: Python, Java, JavaScript, C, CUDA, Dart, Swift, Kotlin C++, HTML/CSS, Verilog, VHDL, ARM, MIPS, RISC-V, x86

Frameworks: React.js, Next.js, Tailwind CSS, Flutter, TensorFlow, React Native, SpringBoot, Ruby on Rails, Django, Pandas

Developer Tools: Firebase, Icarus Verilog, Mbed, OpenCV, OpenGL, Google Cloud, VLSI, Keil μ Vision, Testbenches, Bash, ISIM, OpenMP, Model Sim, Git, Microsoft Azure, Amazon AWS, Oscilloscopes, Arduino, Raspberry Pi, Linux/Unix, SQL